

4-comma: The Artist Robot

ELNAZ SHADKAMI, Texas A&M University, USA



Fig. 1. Sketch AI 4-comma:AI restores a child's creativity

Author's Contact Information: Elnaz Shadkami, Shadkamielnaz@tamu.edu, Texas A&M University, College Station, Texas, USA.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

Conference acronym 'XX, College Station, TX

This study looks at how AI helps Yonkoma storytelling. A child's damaged drawing is fixed using DALL-E 3 and Visual ChatGPT. AI is not only for automation but also supports creativity. It makes artistic work easier, helps beginners, and improves storytelling. Future research will develop AI for even better results.

CCS Concepts: • **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Additional Key Words and Phrases: Do, Not, Us, This, Code, Put, the, Correct, Terms, for, Your, Paper

ACM Reference Format:

Elnaz Shadkami. 2025. 4-comma: The Artist Robot. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 4 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction and Related Works

AI is changing how Yonkoma (4-Koma) manga is made by mixing old storytelling with new technology. Yonkoma manga started in Japan in the early 20th century and tells short but strong stories in four panels. At first, people drew it by hand, but later, it became digital. Now, AI tools help improve the process. Advanced AI models like DALL-E 3 create high-quality manga images from text[1]. This ensures all panels have the same artistic style. Also, Visual ChatGPT lets people modify AI-made images by giving feedback and making adjustments[2]. AI makes manga production faster, more flexible, and more accurate. However, problems remain, such as keeping a stable visual style and making sure images fit the story. To fix this, people save AI settings and structured prompts to get consistent results[3]. This study explores how AI is changing Yonkoma manga. AI automates repetitive tasks, allowing people to focus more on storytelling. This helps them be more creative while keeping the manga's unique style. As AI improves, it will enhance efficiency and artistic expression. AI will play a bigger role in digital manga and change how it is created in the future. Using AI instead of old methods will redefine manga creation.

2 Methodology

This project used AI to create a Yonkoma (4-panel) manga. The goal was to keep the story and images consistent and make the process easy to repeat. The story was divided into four panels, each showing a key moment. A short sentence was written to help AI generate accurate images for each scene. To ensure consistency, prompts included character expressions (sad, happy), objects (glass, spilled water, painting tools), actions (explaining, recreating), and the robot's appearance (a screen on its chest). Prompts were tested and refined multiple times to improve results. Images were generated using DALL-E 3, but sometimes AI made mistakes. To fix this, prompts were adjusted, Visual ChatGPT was used for corrections like expressions and lighting, and sometimes, manual editing was needed. Challenges included keeping character details consistent, ensuring small objects like the glass or the girl's hair color appeared correctly, and maintaining background and lighting continuity. These issues were solved using reference images, refined prompts, and use photoshop. AI was chosen instead of hand drawing because it was faster, more flexible, and allowed for easy modifications. Structured prompts ensured others could follow the same process and achieve similar results, making AI a strong alternative to traditional illustration methods.

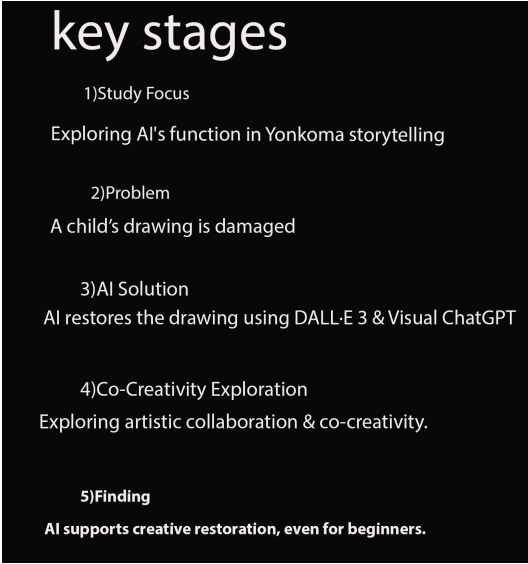


Fig. 2. key Stages

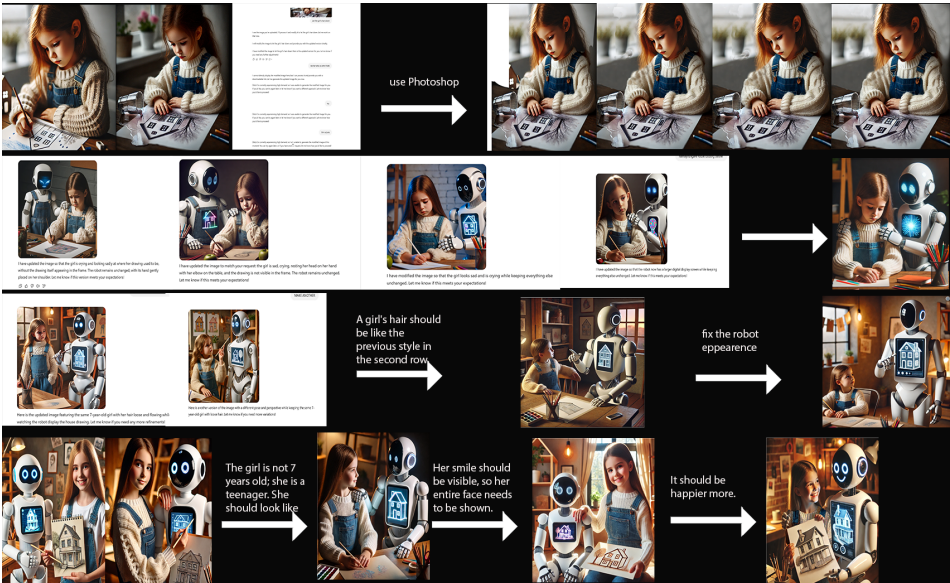


Fig. 3. From structured descriptions to refined images—DALL-E 3 generates, Visual ChatGPT perfects and Photoshop.

3 Result and Future Work

this project studied how AI can help make Yonkoma (4-panel) manga, focusing on automatic storytelling and keeping images consistent. The child means human creativity, and the robot means AI and technology, showing how they can work together to make better stories. AI helped make the story clearer, but keeping the same style in all panels was challenging. Changing the prompts

fixed some problems, but manual editing was still needed. People said the pictures were interesting but not very emotional. In the future, the focus will be on stronger feelings and different styles, leading to a short film using AI tools.

4 Conclusion

This study shows how AI changes Yonkoma manga by improving storytelling while maintaining the art style. It proves AI aids creativity, but challenges remain, such as ensuring image uniformity. In the future, AI should adapt to various styles, improving creativity while preserving the artist's unique touch, balancing automation and expression.

Acknowledgments

This work is submitted as part of Assignment 1 for the VIZA 626 course at Texas A&M University, under the instruction of Professor You-Jin Kim, during the Spring 2025 semester.

[1] Betker, James, et al. "Improving Image Generation with Better Captions." arXiv preprint arXiv:2309.12345, 2023, <https://arxiv.org/abs/2309.12345>. Accessed 28 Feb. 2025.

[2] Wu, C., et al. "Visual ChatGPT: Talking, Drawing, and Editing with Visual Foundation Models." Microsoft Research Asia, 2023, <https://arxiv.org/abs/2303.04671>.

[3] Liu, V., et al. "3DALL-E: Integrating Text-to-Image AI in 3D Design Workflows." Autodesk Research, 2023, <https://doi.org/10.1145/3563657.3596098>.